Mental Health Signal Analysis System:

Evolution from Disorder Detection to Multi-Task NLP Pipeline

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report on "Mental Health Signal Analysis System: Evolution from Disorder Detection to Multi-Task NLP Pipeline" is the bonafide work of "Avi Ambastha", "Ekam Bitt", "Abhishek Prasad", and "Ayush Chhajer who carried out the project.

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ABSTRACT

The exponential growth of social media has created an unprecedented volume of user-generated text that carries latent signals about mental health and emotional well-being. This project presents the Mental Health Signal Analysis System, a privacy-preserving natural language processing platform designed to detect and analyze mental-health-related language patterns in online comments. The system is explicitly framed as an awareness and signal analysis tool and does not claim to provide clinical diagnoses or replace professional mental health evaluation.

The project is organized as a coherent two-version evolution. Version 1 comprises a fully functional Chrome extension that automatically extracts visible comments from supported social media platforms including YouTube, Reddit, and X/Twitter, and routes them to a locally hosted Flask inference backend. The backend employs a fine-tuned RoBERTa model to classify each comment into one of seven categories: ADHD, Anxiety, Autism, Borderline Personality Disorder (BPD), Depression, Post-Traumatic Stress Disorder (PTSD), and Normal or Casual Conversation. All inference occurs entirely on the user's local machine, preserving user privacy. The extension popup presents interactive visualizations including category-wise comment counts, confidence-ranked comment lists, and Chart.js-powered distribution charts. A complementary website interface deployed at https://signal-nine-phi.vercel.app/ provides direct text-based analysis for users who prefer a browser-accessible surface without the extension.

Version 2 introduces a methodological redesign motivated by the scientific and ethical limitations of single-label disorder classification. The revised system shifts from categorical disorder detection to multi-label mental health signal analysis, allowing a single piece of text to simultaneously carry multiple overlapping psychological indicators, each accompanied by a severity estimate. The V2 pipeline maintains the RoBERTa model family as its backbone, incorporates ONNX Runtime quantization to reduce the model deployment footprint from approximately 499 MB to 125 MB, and introduces a SQLite-based local wellbeing history store for optional longitudinal trend visualization.

The fine-tuned RoBERTa V1 classifier achieved an accuracy of 85.69% and a macro F1-score of 0.8601, significantly outperforming BERT-base (79.50%), SVM with TF-IDF features (68.15%), and lexicon-based baselines (38.00%). The project demonstrates that contextual transformer representations are substantially superior to classical approaches for mental health language classification on social media corpora.

The system integrates responsible AI principles throughout its design: all processing remains local, outputs are presented as probabilistic signals rather than diagnoses, and the V2 framing reflects awareness of the ethical limitations of weakly supervised disorder labels derived from subreddit membership. The project contributes a deployable, open-source NLP platform that bridges the gap between academic mental health NLP research and practical, privacy-respecting user-facing systems.

Keywords: *mental health NLP, RoBERTa, social media analysis, browser extension, privacy-preserving inference, multi-label classification, signal analysis, transformer models, ONNX quantization, responsible AI, Flask, Chrome extension, Manifest V3*

LIST OF ABBREVIATIONS

NLP	Natural Language Processing
AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
BERT	Bidirectional Encoder Representations from Transformers
RoBERTa	Robustly Optimized BERT Pretraining Approach
DistillBERT	Destilled BERT
LLM	Large Language Model
ONNX	Open Neural Network Exchange
API	Application Programming Interface
Flask	Python Micro Web Framework
CORS	Cross-Origin Resource Sharing
JSON	JavaScript Object Notation
DOM	Document Object Model
ADHD	Attention Deficit Hyperactivity Disorder
BPD	Borderline Personality Disorder
PTSD	Post-Traumatic Stress Disorder
CPU	Central Processing Unit
GPU	Graphics Processing Unit
INT8	8-bit Integer (quantization format)
SQLite	Self-contained SQL Database Engine
CSV	Comma-Separated Values
TF-IDF	Term Frequency–Inverse Document Frequency
SVM	Support Vector Machine
F1	F1-Score (harmonic mean of Precision and Recall)
MV3	Manifest Version 3 (Chrome Extension Standard)
UI	User Interface
HTTP	HyperText Transfer Protocol
HTTPS	HyperText Transfer Protocol Secure
V1	Version 1
V2	Version 2

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Chapter 1: Introduction

1.1 Background

The proliferation of social media platforms over the past two decades has fundamentally transformed the way individuals express their thoughts, emotions, and personal experiences. Platforms such as YouTube, Reddit, and X (formerly Twitter) collectively host billions of user-generated comments daily, many of which contain candid and emotionally charged language that reflects the psychological states of their authors. Unlike curated clinical interviews or structured mental health questionnaires, these publicly shared expressions offer unfiltered insight into the emotional landscape of digital communities.

Mental health disorders such as depression, anxiety, post-traumatic stress disorder (PTSD), attention deficit hyperactivity disorder (ADHD), autism spectrum conditions, and borderline personality disorder (BPD) affect hundreds of millions of people globally. The World Health Organization estimates that one in eight people worldwide live with a mental disorder, yet access to professional mental health support remains severely limited by financial, geographic, cultural, and stigma-related barriers. In this context, automated computational methods that can identify mental-health-related signals in natural language have attracted significant attention from researchers across computer science, clinical psychology, and public health.

Early work by Coppersmith, Dredze, and Harman [7] demonstrated that social media language contains measurable psychological signals and that computational methods can extract them with meaningful accuracy. Subsequent work by Tadesse et al. [8] showed that Reddit in particular, with its disorder-specific communities (subreddits), provides a rich source of text correlated with mental health experiences. The availability of transformer-based language models such as BERT [2] and RoBERTa [3] has since enabled substantially more accurate and contextually sensitive classification of such text.

However, despite these advances, the majority of academic work in mental health NLP remains confined to research notebooks and benchmark evaluations. There is a significant gap between laboratory-level classification models and deployable systems that can actually be used by individuals in real-world browsing contexts while preserving their privacy. This project addresses that gap directly.

1.2 Motivation

The immediate motivation for this project arose from a simple but practically significant observation: when a user watches a video or reads a post on a social media platform, they are exposed to hundreds or thousands of comments from other users. The emotional and psychological content of that comment stream has a documented influence on viewer mood and mental well-being. Yet there exists no practical tool that allows an ordinary user to understand the aggregate psychological character of the comment environment they are browsing, let alone to detect signals of distress within it.

From a technical standpoint, the motivation was to bridge the gap between the powerful contextual representation capabilities of modern transformer models and the practical needs of a browser-based, privacy-respecting user interface. Existing mental health classification research typically relies on server-side processing, which raises significant privacy concerns when applied to sensitive personal or social data. A local inference model running on the user's own hardware eliminates this concern entirely.

From an ethical standpoint, the team was motivated to build a system that provides genuine value — in the form of signal awareness — without overclaiming. The language of disorder diagnosis is inappropriate for a text classification system trained on social media data. The system is therefore framed throughout as a signal analysis and awareness tool, consistent with the responsible AI principles advocated by Mireshghallah et al. [15] and broader ethics literature on mental health AI [16].

1.3 Problem Statement

Contemporary social media platforms generate comment streams of such volume that no individual user can meaningfully assess the aggregate emotional or psychological character of the conversations they participate in or observe. Existing tools for content moderation and sentiment analysis focus primarily on toxicity and polarity, and are not designed to identify nuanced mental-health-related language patterns. Furthermore, the sensitivity of mental health data makes any cloud-based processing of such content ethically problematic.

The specific problems addressed by this project are therefore: (1) the absence of a practical, locally executed tool for analyzing the mental-health-related signal content of social media comments; (2) the limitations of single-label disorder classification approaches, which fail to capture the overlapping and multi-dimensional nature of emotional language; and (3) the ethical deficit of systems that frame probabilistic text classification as clinical diagnosis.

1.4 Proposed Solution Overview

The proposed solution is a two-component, privacy-preserving mental health signal analysis system comprising a Chrome browser extension and a complementary website interface. The browser extension integrates with supported social media platforms and automatically extracts visible comments for analysis. Comments are transmitted to a locally hosted Flask inference backend, which employs a fine-tuned RoBERTa model to classify each comment into one of seven mental-health-related categories. All inference takes place on the user's local machine; no comment text is transmitted to any external server.

The extension popup presents analysis results through interactive Chart.js visualizations, category-wise counts, confidence-ranked comment lists, and category filters. The website interface, deployed at https://signal-nine-phi.vercel.app/, provides an alternative entry point for users who wish to analyze text directly without relying on the extension. Both surfaces are complementary: the extension provides live, in-context analysis during browsing, while the website provides a cleaner, more accessible standalone analysis experience.

The system design was informed by prior work on privacy-preserving NLP [15] and by the practical architecture patterns described in the Chrome Extensions Manifest V3 developer documentation [18]. The choice of RoBERTa as the classifier backbone was motivated by its demonstrated superiority over earlier encoder architectures for text classification tasks [3], and by its strong performance on social media mental health datasets in prior literature [10].

1.5 System Evolution: Version 1 to Version 2

The project exists as a coherent two-version evolution rather than two disconnected efforts. Version 1 is a fully implemented and deployed system that proves the team's capacity to build, evaluate, and ship a complete end-to-end NLP product. It demonstrates real browser integration, local inference, Docker-based deployment, and quantitative evaluation.

Version 2 represents a methodologically significant redesign motivated by the identified limitations of V1's single-label, subreddit-supervised approach. The V2 system reframes the core task as multi-label mental health signal analysis, introducing severity estimation, improved annotation methodology using local LLM assistance, ONNX Runtime quantization for efficient deployment, and a more mature set of user-facing wellbeing features. Importantly, V2 maintains the RoBERTa model family as its backbone throughout, preserving consistency with V1 and confirming that the project's final deployment posture is RoBERTa-centered.

This evolution from disorder detection to signal analysis is not merely a change of label. It reflects a genuine critical assessment of the ethical and scientific adequacy of the original framing, and the transition is grounded in the responsible AI and mental health ethics literature [16, 17].

1.6 Outline of This Report

The remainder of this report is organized as follows. Chapter 2 presents the objectives of the project. Chapter 3 defines the scope of the system. Chapter 4 provides a thematically organized literature survey covering transformer foundations, mental health NLP, weak supervision, multi-label modelling, efficient deployment, and privacy and ethics. Chapter 5 describes the technology stack. Chapter 6 presents the system architecture and design for both Version 1 and Version 2. Chapter 7 defines the functional requirements. Chapter 8 provides detailed implementation notes. Chapter 9 presents results and discussion including comparative evaluation. Chapter 10 addresses ethical considerations and privacy design. Chapter 11 outlines future scope, and Chapter 12 concludes the report. References are listed at the end.

Chapter 2: Objectives

The Mental Health Signal Analysis System was designed and implemented with a clearly defined set of objectives that collectively span technical implementation, scientific evaluation, ethical responsibility, and user experience. These objectives guided every architectural and design decision made throughout the project and provide the criteria against which the system's success is measured.

2.1 Primary Objectives

2.1.1 To Design and Implement a Privacy-Preserving Mental Health Signal Analysis System

The foremost objective was to build a system capable of identifying mental-health-related language patterns in social media comments while ensuring that all inference occurs locally on the user's device. This objective rules out any cloud-based processing of user-generated text, and it shaped the entire backend and deployment architecture of the project.

2.1.2 To Develop a Functional Chrome Browser Extension with Live Comment Analysis

A core deliverable of the project was a working Chrome extension conforming to the Manifest V3 standard. The extension was required to support comment extraction from at least three major social media platforms (YouTube, Reddit, and X/Twitter), to route extracted comments to the local inference backend, and to present analysis results in an interactive popup interface with visualizations.

2.1.3 To Train and Evaluate a Fine-Tuned RoBERTa Classifier for Seven Mental Health Categories

The project required the training of a RoBERTa-based text classifier capable of categorizing social media comments into seven classes: ADHD, Anxiety, Autism, BPD, Depression, PTSD, and Normal or Casual Conversation. The model was required to be evaluated rigorously against multiple baselines including BERT-base, SVM with TF-IDF features, and a lexicon-based approach.

2.1.4 To Deploy a Complementary Website Interface for Direct Text Analysis

In addition to the browser extension, the project required the development and deployment of a website-based interface that allows users to submit text directly for analysis. This objective was met through the deployment of the website at https://signal-nine-phi.vercel.app/ using

Vercel hosting, with the website repository maintained at https://github.com/achhajer/mental-health-analyzer.

2.2 Secondary Objectives

2.2.1 To Identify and Articulate the Limitations of Single-Label Disorder Classification

An important secondary objective was to conduct a critical analysis of the V1 system's conceptual and empirical limitations. Specifically, the project aimed to demonstrate that single-label classification of mental health language is a simplification of how emotional expression actually works, and that this simplification has both scientific and ethical consequences.

2.2.2 To Design a Version 2 System with Multi-Label Signal Analysis and Severity Estimation

Based on the limitations identified in V1, a secondary objective was to design and partially implement a V2 system that supports multi-label output, where each comment can be assigned multiple concurrent signal labels, each with an associated severity estimate. This redesign also required a shift from subreddit-based weak supervision to a more structured annotation methodology supported by local LLM assistance.

2.2.3 To Apply ONNX Runtime Quantization for Efficient Local Deployment

A practical objective of the V2 pipeline was to reduce the inference model's deployment footprint through ONNX conversion and INT8 quantization, bringing the model size from approximately 499 MB to approximately 125 MB. This objective was motivated by the need to make local inference feasible on consumer hardware without GPU acceleration.

2.2.4 To Maintain Ethical Framing Throughout the System

Across both versions, a consistent objective was to ensure that all system outputs, interface language, and documentation used responsible, non-diagnostic framing. The system was designed to present its outputs as probabilistic signals and awareness indicators, never as clinical assessments or definitive diagnoses. This objective was directly informed by the ethics and responsible AI literature [16, 17].

Chapter 3: Scope of the Project

The scope of the Mental Health Signal Analysis System is defined along multiple axes: the types of content analyzed, the platforms supported, the categories of signals detected, the deployment environments targeted, the user surfaces provided, and the ethical boundaries respected. Understanding the scope helps distinguish what the system is designed to accomplish from what it explicitly does not claim to do.

3.1 Scope of Content Analysis

The system is designed to analyze short-to-medium length text passages of the kind commonly found in social media comment sections. This includes comments on video content platforms such as YouTube, discussion threads on Reddit, and short posts and replies on X/Twitter. The analysis is performed at the level of individual comments, with aggregated summaries computed across the complete set of visible comments on a given page.

The system does not attempt to analyze audio, video, images, or any modality other than text in its current implementation. It does not track individual users across sessions, does not perform any form of user profiling, and does not store any analyzed content beyond the current browser session. In Version 2, a local SQLite wellbeing store is introduced for optional trend tracking, but this operates entirely on the user's own device.

3.2 Scope of Platform Support

The Version 1 extension supports comment extraction from YouTube, Reddit, and X/Twitter. Each platform uses a different DOM structure for rendering comment content, and the extension's content script layer implements platform-specific CSS selectors and Mutation Observer logic to handle each site's dynamic comment loading patterns. The website frontend is platform-agnostic and accepts any text input supplied by the user directly.

Support for additional platforms such as Facebook, Instagram, LinkedIn, or others is considered future scope. The current platform set was selected based on the volume of publicly visible user-generated comments and the suitability of those comments for mental-health-related language analysis.

3.3 Scope of Signal Categories

Version 1 of the system classifies each comment into one of seven predefined categories: ADHD, Anxiety, Autism, BPD, Depression, PTSD, and Normal or Casual Conversation. These categories were selected based on the availability of labeled social media data and the prevalence of community discussion around these conditions on Reddit-based corpora.

Version 2 expands this to a multi-label framework, where a single comment may be associated with multiple signal dimensions simultaneously. The V2 system also introduces severity estimation, providing a qualitative scale (such as low, moderate, and high intensity) for each detected signal. This multi-label, severity-aware output more accurately reflects the complexity of real emotional language.

3.4 Scope of Deployment

The system is designed for local deployment on the user's personal computer. The Flask inference backend runs on localhost and is managed through Docker Compose, allowing the entire backend stack to be launched with a single command. The RoBERTa models are hosted on Hugging Face to facilitate easy download and reproducibility, but inference itself occurs locally once the model has been downloaded.

The website frontend is hosted on Vercel and is accessible globally. However, the backend inference for the website also connects to locally hosted or Hugging Face-served model endpoints, preserving the local-first design philosophy wherever technically feasible.

3.5 Ethical Boundaries and Disclaimers

The system explicitly does not claim to provide clinical diagnoses, psychiatric assessments, or treatment recommendations. It does not replace professional mental health evaluation and is not intended for use in clinical settings. All outputs are presented as probabilistic signal indicators derived from NLP classification, and the interface language reflects this at every point.

The system analyzes publicly available social media comments, not private communications. It does not have access to any data that the user has not explicitly provided or that is not already visible in their browser. No user data is transmitted to any third-party server. This framing aligns with the privacy-preserving NLP principles described by Mireshghallah et al. [15] and the ethical guidelines for mental health AI discussed in [16].

Chapter 4: Literature Survey

A substantial body of research underpins the design and methodology of the Mental Health Signal Analysis System. This chapter surveys the relevant literature organized into six thematic areas: transformer foundations, mental health NLP on social media, weak supervision and dataset quality, multi-label classification and severity modelling, efficient deployment and model distillation, and privacy and ethics in mental health AI. The survey concludes with a gap analysis that identifies the specific contributions of the present work.

4.1 Transformer Foundations

4.1.1 The Transformer Architecture

The foundational architectural innovation underlying all modern encoder-based models used in this project is the self-attention mechanism introduced by Vaswani et al. [1] in their seminal 2017 paper "Attention Is All You Need." The Transformer architecture dispenses entirely with the recurrent and convolutional components that had dominated sequence modelling up to that point, replacing them with a purely attention-based encoder-decoder framework. The key insight is that every token in a sequence can attend to every other token simultaneously, allowing the model to capture both long-range dependencies and local contextual relationships without the sequential bottleneck of recurrent networks. This architectural property is particularly valuable for mental health language analysis, where meaning is often conveyed through subtle cues distributed across an entire comment rather than in isolated keywords.

4.1.2 BERT and Masked Language Modelling

Devlin et al. [2] extended the Transformer encoder into a general-purpose pre-trained language representation system through BERT (Bidirectional Encoder Representations from Transformers). BERT's key innovation was the masked language modelling (MLM) pre-training objective, which trains the model to predict randomly masked tokens from their surrounding context in both directions simultaneously. This bidirectional contextualization produces representations that are substantially richer than the unidirectional embeddings produced by earlier models such as ELMo or GPT. BERT established the fine-tuning paradigm that this project employs: a large model pre-trained on general corpora is subsequently fine-tuned on a smaller, task-specific labeled dataset. BERT-base serves as a comparison baseline in the Version 1 evaluation of this project.

4.1.3 RoBERTa: Robustly Optimized BERT Pretraining

Liu et al. [3] demonstrated that BERT had been systematically undertrained and that key modifications to the pre-training procedure could yield significantly improved representations without altering the underlying architecture. RoBERTa (Robustly Optimized BERT Pretraining Approach) removes the next-sentence prediction objective, trains on larger batches with longer sequences, uses dynamic masking rather than static masking, and trains on substantially more data than the original BERT. These changes produce a model that outperforms BERT across a wide range of NLP benchmarks. RoBERTa is the primary classifier backbone used in Version 1 of this project and remains the model family of choice throughout Version 2. Its demonstrated superiority for text classification tasks and its established community of fine-tuned domain-specific variants made it the natural choice for a mental health language analysis system.

4.1.4 Domain-Adapted RoBERTa Variants

The development of domain-specific RoBERTa variants, including models pre-trained or fine-tuned specifically on mental health-related social media text, has further expanded the applicability of RoBERTa to this domain [4]. Such domain adaptation, whether achieved through continued pre-training on in-domain corpora or through task-specific fine-tuning, consistently improves performance on mental health NLP tasks compared to general-purpose checkpoints. The project's use of a fine-tuned RoBERTa model trained on mental health subreddit data is consistent with this established direction in the literature.

4.2 Mental Health NLP on Social Media

4.2.1 Early Signal Detection Work

Coppersmith, Dredze, and Harman [7] conducted pioneering work demonstrating that language use on Twitter contains statistically detectable signals correlated with mental health conditions including post-traumatic stress disorder, depression, bipolar disorder, and seasonal affective disorder. By constructing datasets from users who self-disclosed diagnoses in their profiles and comparing their language patterns to control users, the authors showed that computational linguistic features could meaningfully distinguish between these groups. This early work established the fundamental premise of the present project: that social media language carries genuine psychological signal.

4.2.2 Reddit-Based Depression Detection

Tadesse et al. [8] extended mental health signal detection to the Reddit platform, demonstrating that posts from depression-related subreddits exhibit distinctive linguistic characteristics that machine learning classifiers can learn to identify. Their work used a combination of LIWC features, n-gram features, and several classical classifiers to detect depression-related posts from the r/depression subreddit compared to control posts from r/casual. The work is directly relevant to this project because the Version 1 dataset was similarly constructed from Reddit subreddit communities corresponding to the seven target categories.

4.2.3 CLPsych and Shared Task Benchmarks

The CLPsych (Computational Linguistics and Clinical Psychology) shared tasks [9] have produced a series of benchmark datasets and challenge problems focused on suicide risk assessment, depression detection, and mental health status prediction from social media text. These challenges have driven the development of increasingly sophisticated NLP approaches and have helped establish community standards for evaluation methodology, ethical data handling, and responsible framing of model outputs. The existence of these benchmarks confirms that mental health NLP is a mature and well-studied research direction with established evaluation protocols.

4.2.4 Transformer-Era Mental Health Classification

More recent work [10] has applied BERT and RoBERTa directly to mental health classification tasks on Reddit and Twitter corpora, consistently showing that transformer-based representations outperform classical feature engineering approaches by substantial margins. These papers bridge the gap between the early signal detection literature and the current state of the art, confirming that contextual embeddings from pre-trained transformers are the appropriate tool for this domain. The Version 1 evaluation in this project reproduces this finding empirically, with the fine-tuned RoBERTa model achieving 85.69% accuracy against BERT-base at 79.50% and SVM/TF-IDF at 68.15%.

4.3 Weak Supervision and Dataset Quality

4.3.1 Subreddit-Based Weak Supervision

A common methodology in mental health NLP dataset construction involves assigning labels to text samples based on the subreddit from which they were collected [11]. Posts from r/depression are labeled as depression-positive; posts from r/ADHD are labeled as ADHD-positive, and so on. While this approach is pragmatically convenient and produces large labeled datasets without expensive manual annotation, it introduces systematic label noise. A user posting in r/depression may be seeking general support rather than expressing depressive symptoms; a user posting in r/ADHD may be a parent or clinician rather than someone with the condition. This subreddit membership heuristic constitutes a form of weak supervision that the Version 2 redesign explicitly addresses.

4.3.2 Label Noise in Social Media NLP

The broader literature on label noise in clinical and social media NLP [12] documents the ways in which proxy labels and crowdsourced annotations can systematically mislabel instances, degrade model calibration, and produce models that learn surface-level associations rather than genuine signal. For mental health classification specifically, label noise is compounded by the inherent ambiguity of emotional language and the absence of ground-truth clinical labels for training. The Version 2 redesign uses LLM-assisted annotation with local tools to generate more structured and nuanced labels, reducing dependence on subreddit membership as the sole source of supervision.

4.4 Multi-Label Classification and Severity Modelling

4.4.1 Multi-Label Text Classification

Single-label classification of emotional or psychological text is a simplification of how human language actually works. A single comment may simultaneously express anxiety about the future, grief about the past, and social withdrawal — none of which can be adequately captured by assigning a single disorder label. Multi-label text classification methods [13] allow a text sample to be associated with multiple labels simultaneously, better reflecting the co-occurring and overlapping nature of emotional states. The Version 2 shift to multi-label signal analysis is grounded in this literature and is presented as a scientific improvement over the V1 single-label approach.

4.4.2 Ordinal and Severity Prediction

In addition to binary or multi-class prediction, there is a growing body of work on ordinal classification and severity estimation in clinical NLP [14]. Severity is not simply another label; it is an ordered property that requires specialized modelling approaches such as ordinal regression or hierarchical label structures. The Version 2 system incorporates severity estimation for each detected signal, providing users with a more informative output than a simple category label. This is consistent with the literature's emphasis on moving beyond presence/absence detection toward severity-aware analysis.

4.5 Efficient Deployment and Model Distillation

4.5.1 Knowledge Distillation

Hinton, Vinyals, and Dean [5] introduced knowledge distillation as a method for compressing the knowledge of a large, high-performing teacher model into a smaller, more efficient student model. The student is trained not only on hard ground-truth labels but also on the soft probability distributions output by the teacher, which carry richer information about inter-class relationships. The Version 2 pipeline employs a teacher–student training paradigm in which a full-scale RoBERTa teacher model supervises the training of a more compact student model suitable for local deployment.

4.5.2 ONNX Runtime and Quantization

The Open Neural Network Exchange (ONNX) format and the associated ONNX Runtime [6, 19] provide a hardware-agnostic inference engine that supports model optimization techniques including INT8 quantization. By converting the student model to ONNX and applying post-training quantization, the Version 2 deployment pipeline reduces the model size from approximately 499 MB (the full V1 RoBERTa model) to approximately 125 MB, enabling practical CPU-based local inference on consumer hardware without GPU acceleration. This deployment strategy is essential for the project's local-first privacy philosophy.

4.6 Privacy and Ethics in Mental Health AI

4.6.1 Privacy-Preserving NLP

Mireshghallah et al. [15] survey the landscape of privacy-preserving NLP, covering approaches including federated learning, differential privacy, local inference, and data anonymization. The most directly relevant approach for this project is local inference, which ensures that sensitive text data never leaves the user's device. By hosting the inference backend on localhost rather than a remote server, the Mental Health Signal Analysis System eliminates the network transmission of user data entirely, providing the strongest possible privacy guarantee within the constraints of a practical deployed system.

4.6.2 Ethical Considerations in Mental Health AI

A growing body of literature [16] addresses the ethical challenges specific to machine learning systems applied to mental health contexts. Key concerns include the risk of stigmatization through automated labeling, the absence of informed consent from users whose public posts are used as training data, the potential for false positives to cause unwarranted alarm, and the inappropriate use of probabilistic outputs as clinical facts. This project addresses these concerns through its consistent use of signal analysis terminology rather than diagnostic language, its transparent presentation of outputs as probabilistic estimates, and its explicit framing as an awareness tool rather than a clinical instrument.

4.6.3 Explainability and Safe Interface Design

Work on explainability in mental health AI [17] emphasizes that systems operating in sensitive domains should provide users with meaningful explanations of their outputs and should be designed to avoid creating undue alarm or dependency. The Version 2 system's user-facing features, including Smart Shield content filters, Rabbit-Hole Nudges for prolonged high-distress browsing sessions, and Draft-Time Support Prompts before users post potentially distress-indicating content, are all designed with these principles in mind. These features extend the system from passive signal detection to active, ethically informed user support.

4.7 Gap Analysis

The literature review reveals several gaps that this project directly addresses. First, while academic research in mental health NLP has demonstrated strong classification performance, there is a notable absence of deployable, locally executing, user-facing systems that bring these capabilities to ordinary users in their real browsing environments. Second, the dominant subreddit-based weak supervision methodology in dataset construction has known limitations that have not been systematically addressed by existing deployment-oriented work. Third, the single-label classification paradigm remains prevalent even though it is scientifically and ethically inferior to multi-label signal analysis for this domain.

This project addresses all three gaps: it delivers a fully deployed local system, it identifies and redesigns around the limitations of weak supervision in V2, and it introduces multi-label signal analysis with severity estimation as a more principled approach. These contributions collectively position the project at the intersection of mental health NLP research and practical, responsible AI development.

Chapter 5: Literature Survey

The Mental Health Signal Analysis System was built using a carefully selected stack of open-source technologies spanning frontend development, backend inference, machine learning, and infrastructure. Each technology choice was driven by specific technical requirements and is described below.

5.1 Frontend Technologies

5.1.1 Chrome Extension with Manifest V3

The browser extension is built using the Chrome Extensions Manifest V3 standard [18], which represents the current generation of the Chrome extension platform. Manifest V3 introduces a service-worker-based background architecture, replacing the persistent background page model of Manifest V2. This architecture is more memory-efficient and aligns with modern browser security models. The extension consists of three principal JavaScript components: a content script injected into supported web pages for comment extraction, a background service worker that manages state, batching, and communication between the content script and the popup, and the popup interface rendered in a separate browser context.

5.1.2 HTML, CSS, and JavaScript

The extension popup interface is implemented in vanilla HTML, CSS, and JavaScript, without any external frontend framework dependencies. This choice keeps the extension lightweight and avoids the complexity of bundling a framework into the extension package. The popup communicates with the background service worker using the Chrome Extensions messaging API (chrome.runtime.sendMessage and chrome.runtime.onMessage), and accesses cached state through chrome.storage.local.

5.1.3 Chart.js

Data visualizations within the extension popup are rendered using Chart.js, a widely adopted open-source JavaScript charting library. Chart.js provides interactive, responsive charts including bar charts, doughnut charts, and pie charts, which are used to display the category-wise distribution of detected mental health signals across the analyzed comment set. Chart.js was selected for its ease of integration with vanilla JavaScript, its extensive chart type library, and its smooth animation capabilities.

5.1.4 Website Frontend

The complementary website interface at https://signal-nine-phi.vercel.app/ provides a browser-accessible surface for direct text analysis. The website frontend, maintained in the repository at https://github.com/achhajer/mental-health-analyzer, accepts free-form text input from the user, submits it to the RoBERTa-backed inference endpoint, and displays the analysis results in a clean, user-friendly layout. The website is deployed on Vercel, providing global CDN distribution and continuous deployment from the GitHub repository.

5.2 Backend Technologies

5.2.1 Python 3.10

The inference backend is written in Python 3.10, which provides strong library support for NLP, machine learning, and web service development. Python 3.10's structural pattern matching and improved type hinting features were utilized during development for cleaner code organization.

5.2.2 Flask and Flask-CORS

The web framework serving the inference API is Flask, a lightweight Python micro-framework well suited to single-purpose API services. Flask's simplicity and low overhead make it ideal for an inference server that handles a single endpoint type: receiving a batch of comments and returning a list of classification predictions. Flask-CORS is used to enable cross-origin resource sharing, allowing the Chrome extension's background service worker to make HTTP requests to the localhost Flask server without browser security policy violations.

5.2.3 Gunicorn

In production-mode local deployment, the Flask application is served by Gunicorn (Green Unicorn), a Python WSGI HTTP server that provides multi-worker process management. Gunicorn improves throughput for concurrent requests and is the standard production server choice for Flask applications. In the Docker Compose setup, Gunicorn is the entry point for the backend container.

5.3 Machine Learning and NLP Technologies

5.3.1 Hugging Face Transformers

The Hugging Face Transformers library provides the primary interface for loading, fine-tuning, and running inference with RoBERTa and related models. The library's AutoTokenizer and AutoModelForSequenceClassification classes abstract the tokenization and classification pipeline, enabling rapid prototyping and consistent preprocessing across training and inference.

Trained model checkpoints are hosted on the Hugging Face Model Hub for easy download and version management.

5.3.2 PyTorch

PyTorch serves as the underlying deep learning framework for model training and inference. The RoBERTa fine-tuning pipeline uses PyTorch's DataLoader, gradient computation, and optimizer utilities. PyTorch was selected over TensorFlow for its dynamic computation graph, which provides superior debugging capability and greater flexibility during experimental development.

5.3.3 ONNX Runtime

For the Version 2 deployment pipeline, ONNX Runtime [19] is used to convert the trained student model from PyTorch format to the ONNX interchange format and to perform INT8 post-training quantization. The ONNX Runtime inference engine provides hardware-agnostic, highly optimized CPU inference that does not require a GPU, making local deployment on consumer hardware practical. The quantized model achieves a size reduction from approximately 499 MB to approximately 125 MB.

5.4 Infrastructure Technologies

5.4.1 Docker and Docker Compose

The backend deployment stack is containerized using Docker and orchestrated with Docker Compose. A single docker-compose up command starts the Flask/Gunicorn backend container with all required dependencies pre-installed. This containerization approach ensures reproducibility across different operating system environments and eliminates dependency management issues for end users who wish to run the backend locally.

5.4.2 Hugging Face Model Hub

Trained RoBERTa model checkpoints are hosted on the Hugging Face Model Hub, allowing the backend container to download the correct model version at startup using a simple repository identifier. This separation of model artifacts from application code simplifies the deployment container size and enables model versioning without rebuilding the container.

5.4.3 Vercel

The website frontend is deployed on Vercel, a cloud platform specialized for frontend deployment that provides automatic HTTPS, global CDN distribution, and continuous deployment from GitHub. Each push to the main branch of the website repository triggers an automatic Vercel deployment, ensuring that the live site is always in sync with the latest code.

5.4.4 Git and GitHub

Version control is maintained using Git with three primary GitHub repositories: the main extension and V1 backend repository at https://github.com/Ekam-Bitt/Detection-of-Mental-Disorders-Extension, the website frontend repository at https://github.com/achhajer/mental-health-analyzer, and the V2 research and model development repository at https://github.com/Abhishek6792/Final_Year_Project_V2.

Table: Technology Stack Summary

Category	Technology	Purpose
Frontend	HTML, CSS, JavaScript	Extension popup and website UI
Frontend	Chart.js	Interactive visualizations in popup
Frontend	Chrome Extensions MV3	Browser integration and messaging
Backend	Python 3.10	Backend runtime language
Backend	Flask + Flask-CORS	REST API for inference requests
Backend	Gunicorn	Production WSGI HTTP server
ML/NLP	Hugging Face Transformers	RoBERTa model loading and inference
ML/NLP	PyTorch	Model training and fine-tuning
ML/NLP	ONNX Runtime	V2 quantized inference deployment
Infrastructure	Docker + Docker Compose	Containerized local deployment
Infrastructure	Hugging Face Model Hub	Model hosting and versioning
Infrastructure	Vercel	Website frontend hosting and CDN
Storage	SQLite (V2)	Local wellbeing history store
VCS	Git + GitHub	Source code version control

Chapter 6: System Architecture and Design

The Mental Health Signal Analysis System is architected as a modular, local-first platform composed of several interacting components. This chapter describes the system architecture for both Version 1 and Version 2, covering the extension architecture, the Flask inference backend, the model layer, the website frontend, and the data flow across all components.

6.1 Version 1 System Architecture

6.1.1 Architecture Overview

Version 1 is organized as a three-layer system: the browser extension layer, the local inference backend layer, and the model layer. The browser extension handles all user interaction and platform-specific comment extraction. The local Flask backend receives comments, applies preprocessing, and routes them through the RoBERTa model for classification. The model layer itself is a fine-tuned RoBERTa sequence classifier loaded from a Hugging Face-hosted checkpoint.

[Figure 6.1: High-Level System Architecture of the Mental Health Signal Analysis System — insert diagram here]

6.1.2 Chrome Extension Architecture

The extension is composed of three primary components operating within the Chrome extension process model. The content script is injected into the active tab by the extension manifest's content_scripts declaration. It is responsible for detecting the current platform (YouTube, Reddit, or X/Twitter), selecting the appropriate comment extraction strategy, and collecting visible comment text from the DOM.

Comment extraction on dynamic platforms requires more than a one-time DOM query. YouTube and Reddit both load comments asynchronously after the page shell has rendered, and they continuously inject new comments as the user scrolls. The content script uses the browser's MutationObserver API to watch for new comment elements being added to the DOM. When new comments are detected, the observer triggers a debounced rescan of the comment container, ensuring that all visible comments — including those loaded after the initial page render — are captured without generating excessive network requests.

Site-specific CSS selectors are defined for each supported platform, isolating the comment text elements from the surrounding page structure. For YouTube, the selector targets the ytd-comment-renderer elements within the comment section. For Reddit, the comment body elements within the thread view are targeted. For X/Twitter, tweet content elements within thread and reply views are extracted.

Once extracted, comment text is deduplicated using a simple hash-based filter and batched for transmission to the background service worker via chrome.runtime.sendMessage. The background service worker acts as the orchestration hub between the content script, the popup interface, and the Flask backend.

6.1.3 Background Service Worker

The background service worker is a persistent-but-event-driven JavaScript module operating in the extension's background context. It receives batched comment arrays from the content script, maintains a chrome.storage.local cache of recent page analyses to avoid redundant inference calls, and sends HTTP POST requests to the Flask backend's /classify endpoint.

When the backend returns classification results, the background worker updates the stored analysis state and notifies the popup interface if it is open. This decoupled architecture ensures that the popup can be opened and closed without losing analysis state, since all results are persisted in chrome.storage rather than being held in the popup's JavaScript context.

6.1.4 Popup User Interface

The popup interface provides the primary user-facing display for analysis results. It is structured into three panels: a summary panel showing the total comment count, the scan status, and the category-wise comment counts; a visualization panel displaying a Chart.js doughnut or bar chart of the category distribution; and a comment browser panel providing a filterable, scrollable list of analyzed comments with their predicted categories and confidence scores.

The popup initializes by reading cached analysis state from chrome.storage.local and rendering whatever results are available. A manual rescan button allows the user to trigger a fresh analysis of the current page. Category filter buttons allow the user to view only comments associated with a specific signal category, facilitating more focused inspection of the results.

[Figure 6.4: Extension Popup User Interface Layout — insert screenshot here]

6.1.5 Flask Inference Backend

The backend is a single-purpose Flask application exposing one primary endpoint: POST /classify. This endpoint accepts a JSON payload containing an array of comment strings and returns a JSON array of classification results, each containing the predicted category label and the full probability distribution across all seven classes.

The backend applies a minimal preprocessing pipeline to each comment before inference: stripping HTML entities, normalizing whitespace, and truncating comments that exceed the RoBERTa model's maximum token length of 512 tokens. The RoBERTa tokenizer from the Hugging Face Transformers library handles subword tokenization, adding the special [CLS] and [SEP] tokens required by the model.

Comments are processed in mini-batches to balance throughput and memory usage. The model runs in inference mode (torch.no_grad()) to avoid storing gradient computation graphs that are unnecessary during deployment. A Flask-CORS wrapper on the /classify endpoint permits cross-origin requests from the extension's background service worker running in the chrome-extension:// scheme.

[Figure 6.2: Version 1 Data Flow Diagram — Comment Extraction to Inference — insert diagram here]

6.1.6 Dockerized Deployment

The Version 1 backend is packaged as a Docker container with a Dockerfile that installs all Python dependencies, downloads the RoBERTa model checkpoint from Hugging Face, and launches the Gunicorn server on the configured port. A docker-compose.yml file orchestrates this container, mounting the model cache directory as a volume to avoid re-downloading the model on each container restart. Users launch the backend with a single docker-compose up command and the extension is then fully operational.

6.2 Version 2 System Architecture

6.2.1 Redesign Motivation

Version 2 was motivated by three categories of limitation identified in Version 1. First, the single-label classification output is scientifically inadequate for emotional language that simultaneously expresses multiple psychological states. Second, the subreddit-based weak supervision used in V1 dataset construction introduces systematic label noise that limits model reliability. Third, the full-scale RoBERTa model's 499 MB footprint creates deployment challenges for users with limited storage or memory.

6.2.2 Multi-Label Signal Analysis Pipeline

The V2 system reframes the classification task as multi-label signal detection. Each comment is processed through a multi-task RoBERTa-based model that outputs binary detection scores for multiple concurrent signal dimensions, along with severity estimates for each detected signal. The signal taxonomy in V2 is designed to capture overlapping emotional indicators rather than mutually exclusive disorder categories.

[Figure 6.3: Version 2 System Architecture — Multi-Label Signal Analysis Pipeline — insert diagram here]

6.2.3 Teacher–Student Training Architecture

The V2 training pipeline employs a teacher–student knowledge distillation approach [5]. A full-scale RoBERTa teacher model is trained on the improved V2 dataset with multi-label and severity annotations. The teacher's soft output distributions are then used to supervise the training of a more compact student model, which learns not only from the ground-truth labels but also from the richer probabilistic information encoded in the teacher's outputs. This approach preserves much of the teacher's performance while substantially reducing the deployment model's size.

6.2.4 Improved Annotation Methodology

The V2 dataset construction methodology reduces reliance on subreddit membership as the sole source of labels. Instead, a semi-automated annotation pipeline uses a locally deployed large language model (through tools such as Ollama) to review each collected text sample and assign structured multi-label annotations based on explicit signal criteria. This approach produces annotations that are more aligned with the semantic content of the text and less dependent on the community context of the subreddit from which the text was collected.

6.2.5 ONNX Quantization and Local Deployment

Once the student model is trained, it is exported to the ONNX format and subjected to post-training INT8 quantization using the ONNX Runtime quantization tools [19]. This process reduces the model's storage footprint from approximately 499 MB to approximately 125 MB and accelerates inference speed on CPU hardware. The quantized ONNX model is the primary deployment artifact for V2, enabling practical local inference on consumer hardware.

6.2.6 SQLite Wellbeing Store

Version 2 introduces an optional local wellbeing history store implemented using SQLite. When enabled by the user, the extension writes a timestamped record of each page analysis session, including the detected signal distribution and aggregate severity indicators, to a local SQLite database. This store enables longitudinal trend visualization: over time, the user can view how the signal character of their browsing environment has changed, presented through a dashboard-style interface in the extension or website.

6.2.7 User-Facing Wellbeing Features

Version 2 introduces three wellbeing-oriented user interface features that extend the system from passive signal detection to active user support. The Smart Shield feature allows users to configure automatic blurring of comments that exceed a configurable distress signal threshold, reducing involuntary exposure to distress-indicating content. The Rabbit-Hole Nudge feature monitors cumulative high-distress exposure during a browsing session and triggers a gentle awareness notification when the user has been exposed to an elevated signal environment for a prolonged period. The Draft-Time Support Prompt feature activates when the user begins typing a response that the local model identifies as potentially high-distress, offering a reflective pause before posting.

Additionally, V2 introduces an Emotional Diet summary that aggregates the signal character of a browsing session into three broad categories — Calm, Watchful, and Intense — providing users with an intuitive, non-clinical summary of their recent browsing environment. These features are grounded in the responsible design literature [17] and are designed to provide awareness and gentle support rather than clinical intervention.

6.3 Website Frontend Architecture

The website frontend at https://signal-nine-phi.vercel.app/ serves as a complement to the browser extension, providing a standalone entry point for direct text analysis. The website presents a clean text input interface, submits the entered text to the backend inference endpoint, and displays the classification results in a structured output panel.

The website is built and deployed using the GitHub repository at https://github.com/achhajer/mental-health-analyzer. Deployment is managed through Vercel's continuous deployment pipeline, which automatically builds and deploys the latest version of the site on each push to the main branch. The website connects to RoBERTa-based models hosted on Hugging Face, ensuring that inference is powered by the same model family as the extension backend.

The relationship between the extension and the website is complementary rather than redundant. The extension provides in-context, live analysis of comment streams during active browsing, while the website provides a more accessible surface for users who wish to analyze specific text passages without installing the extension. Together, the two surfaces ensure that the system's capabilities are available to the broadest possible audience.

Chapter 7: Functional Requirements

This chapter defines the functional requirements of the Mental Health Signal Analysis System across both Version 1 and Version 2. Requirements are organized by system component and cover input handling, processing, output presentation, user interaction, deployment, and extension-website integration.

7.1 Comment Extraction Requirements

The content script component of the browser extension shall detect when the user is visiting a supported social media platform (YouTube, Reddit, or X/Twitter) and automatically activate the appropriate comment extraction logic. The extraction logic shall use platform-specific CSS selectors to identify comment container elements and shall extract the text content of each visible comment.

The extraction component shall use the MutationObserver API to monitor the page's DOM for newly loaded comment elements and shall trigger a debounced rescan upon detecting new content. This ensures that comments loaded asynchronously after the initial page render are included in the analysis. The component shall deduplicate extracted comments using a content-based hashing strategy to prevent repeated submission of identical text to the backend.

7.2 Backend Inference Requirements

The Flask backend shall expose a POST /classify endpoint that accepts a JSON payload containing an array of comment strings. It shall apply preprocessing including HTML entity stripping, whitespace normalization, and token length truncation before passing comments to the RoBERTa model. The endpoint shall return a JSON array of classification results, each containing the predicted class label and the complete probability distribution across all seven categories.

The backend shall process comments in configurable mini-batches to balance inference throughput and memory utilization. All inference shall be executed on the local machine without transmitting any comment text to external servers. The backend shall respond with appropriate HTTP error codes for malformed requests and shall include descriptive error messages in the response body.

7.3 Extension User Interface Requirements

The extension popup shall display the following information upon opening: the total number of comments analyzed on the current page, the number of comments assigned to each of the seven signal categories, and a visual chart representing the category distribution. The popup shall provide a filterable comment list allowing the user to view only comments associated with a specific category. A manual rescan button shall allow the user to trigger a fresh analysis of the current page.

The popup shall display the current scan status, including whether a scan is in progress, whether the backend is reachable, and whether any errors occurred during extraction or inference. In the absence of backend connectivity, the popup shall display a clear error message rather than an empty or undefined state.

7.4 Website Interface Requirements

The website frontend shall provide a text input area that accepts free-form text of arbitrary length for direct analysis. Upon submission, the website shall display the predicted signal categories and their associated confidence scores. The website shall be accessible via HTTPS at the deployed Vercel URL. The website shall handle backend unavailability gracefully and shall display a user-friendly error message in such cases.

7.5 Version 2 Functional Requirements

The Version 2 system shall support multi-label classification output, where each comment may be assigned one or more concurrent signal labels along with severity estimates for each detected label. The system shall maintain a local SQLite wellbeing store that records timestamped analysis sessions when the feature is enabled by the user. The wellbeing store shall never transmit stored data to any external service.

The Smart Shield feature shall allow the user to configure a distress signal threshold above which comment content is automatically blurred in the browser view. The Rabbit-Hole Nudge feature shall track cumulative high-distress exposure during a session and trigger an awareness notification when the configured threshold is exceeded. The Draft-Time Support Prompt feature shall activate when the user begins composing a reply that the local model identifies as high-distress, offering a brief reflective pause prompt.

The Emotional Diet summary shall present the aggregate signal character of the current browsing session in three categories — Calm, Watchful, and Intense — through the extension popup or website dashboard.

Table: Functional Requirements of the Mental Health Signal Analysis System

Req. ID	Component	Description
FR-01	Content Script	Extract visible comments from YouTube, Reddit, X/Twitter using platform-specific selectors
FR-02	Content Script	Use MutationObserver to capture dynamically loaded comments
FR-03	Background Worker	Batch comments and submit to Flask /classify endpoint via HTTP POST
FR-04	Background Worker	Cache analysis results in chrome.storage.local
FR-05	Flask Backend	Classify comment batches into seven signal categories and return probability distributions
FR-06	Popup UI	Display category counts, distribution chart, and filterable comment list
FR-07	Website	Accept direct text input and return signal analysis results via Hugging Face-hosted model
FR-08	V2 System	Produce multi-label signal output with severity estimates per comment
FR-09	V2 System	Maintain optional local SQLite wellbeing history without external data transmission
FR-10	V2 System	Provide Smart Shield, Rabbit-Hole Nudge, Emotional Diet, and Draft-Time Prompt features

Chapter 8: Implementation Details

This chapter provides a detailed account of the implementation of the Mental Health Signal Analysis System, covering dataset construction, model training, the Chrome extension, the Flask backend, the website frontend, and the Version 2 redesign. The implementation descriptions refer to the completed and deployed system components.

8.1 Dataset Construction and Preprocessing

8.1.1 Data Collection

The Version 1 training dataset was constructed by collecting posts and comments from seven mental-health-related subreddit communities on Reddit: r/ADHD, r/Anxiety, r/autism, r/BPD, r/depression, r/ptsd, and r/CasualConversation (used as the Normal or Casual Conversation class). Reddit's public API was used to collect post titles and body text from each community. Each collected text sample was assigned a class label based on its subreddit of origin, constituting a form of distant supervision or weak labeling [11].

The collection process targeted posts from each subreddit that were substantively authored (i.e., not empty or removed) and that contained sufficient text for meaningful classification (a minimum token count threshold was applied). Duplicate posts, moderator announcements, and bot-generated content were excluded. The final dataset contained samples across all seven categories with approximate class balance maintained through stratified sampling.

8.1.2 Preprocessing Pipeline

Each collected text sample was passed through a standardized preprocessing pipeline before tokenization. The pipeline performed the following operations in sequence: removal of Reddit-specific formatting markers such as markdown emphasis, strikethrough, and blockquote characters; removal of URLs and hyperlinks; stripping of special characters and non-ASCII symbols; normalization of whitespace including collapsing multiple consecutive whitespace characters into a single space; and removal of excessively short samples (fewer than ten words) that are unlikely to contain classifiable content.

The cleaned text was then tokenized using the RoBERTa tokenizer from the Hugging Face Transformers library. The tokenizer applies byte-pair encoding (BPE) to produce subword tokens, prepends the special [CLS] token, and appends the [SEP] token. Sequences exceeding 512 tokens (the maximum sequence length supported by RoBERTa) were truncated from the right, preserving the beginning of each comment which typically contains the most informative content.

8.2 Model Training and Fine-Tuning (Version 1)

8.2.1 Base Model Selection

The base model selected for fine-tuning was the RoBERTa-base checkpoint available through the Hugging Face Model Hub (facebook/roberta-base). RoBERTa was selected on the basis of its demonstrated superiority over BERT across a range of text classification benchmarks [3] and its suitability for social media language, which benefits from the model's dynamic masking strategy and larger pre-training corpus. A classification head consisting of a linear layer with dropout was added on top of the [CLS] token representation to produce the seven-class output.

8.2.2 Training Configuration

The model was fine-tuned using the AdamW optimizer with a learning rate of 2e-5 and a linear warmup schedule over the first 10% of training steps, followed by linear decay to zero. Training was conducted for five epochs with a batch size of 16. A dropout rate of 0.1 was applied to the classification head. The cross-entropy loss function was used, with class weights adjusted to account for any residual class imbalance in the dataset. Model checkpoints were saved at the end of each epoch, and the checkpoint with the highest validation macro F1-score was selected as the final model.

Training was performed on a CUDA-enabled GPU. The trained model was subsequently uploaded to the Hugging Face Model Hub for hosting and was referenced by the backend container using the Hugging Face AutoModel download API.

8.2.3 Baseline Models

Three baseline models were trained and evaluated for comparison: a BERT-base model fine-tuned using the same configuration as the RoBERTa model; an SVM classifier with a linear kernel trained on TF-IDF feature vectors computed from the preprocessed training corpus with a maximum vocabulary size of 50,000 terms and n-gram range of (1, 2); and a lexicon-based classifier using a curated dictionary of mental-health-related terms and sentiment lexicons to assign category labels based on keyword frequency patterns.

[Figure 8.1: RoBERTa Fine-Tuning Pipeline for Seven-Class Classification — insert diagram here]

8.3 Chrome Extension Implementation

8.3.1 Manifest Configuration

The extension manifest (manifest.json) declares the extension's metadata, permissions, content script injection rules, and background service worker. The manifest version is set to 3 (Manifest V3). Permissions declared include storage (for chrome.storage.local access), activeTab (for content script access to the current tab), and scripting (for programmatic content script injection where needed). The host_permissions field includes the URL patterns for the three supported platforms.

8.3.2 Content Script Implementation

The content script (content.js) implements a platform detection function that examines the current page's hostname and URL pattern to determine the active platform. Based on the detected platform, the script selects the appropriate comment container selector from a configuration object that maps platform identifiers to their corresponding DOM query strings.

The extraction function queries the selected container for comment text elements and maps each element to its trimmed text content. A Set-based deduplication filter removes any comments whose normalized text has already been collected in the current session. The MutationObserver is configured to observe the comment container's subtree for childList mutations, triggering a debounced rescan with a 500-millisecond delay to avoid excessive scanning during rapid DOM updates.

Extracted comment arrays are sent to the background service worker using chrome.runtime.sendMessage with a message type identifier of COMMENTS_EXTRACTED. The content script listens for a SCAN_COMPLETE response message from the background worker to confirm successful processing.

[Figure 8.2: Chrome Extension Content Script and Background Worker Interaction — insert diagram here]

8.3.3 Background Service Worker Implementation

The background service worker (background.js) maintains the extension's central state object, which tracks the current page URL, the last analysis timestamp, the analysis results, and any active scan status. The state object is persisted to chrome.storage.local after each update to ensure that popup instances can access current results even when opened after the analysis completes.

Upon receiving a COMMENTS_EXTRACTED message from the content script, the background worker checks whether a scan is already in progress and whether the cached results for the current page URL are still fresh (within a configurable staleness window). If fresh results are available, the cached results are returned immediately. If a fresh scan is required, the worker constructs the HTTP POST request payload, submits it to the Flask backend endpoint at http://localhost:5000/classify, and awaits the response. Error handling covers network failures, backend timeout, and malformed response payloads.

8.3.4 Popup Interface Implementation

The popup (popup.html and popup.js) initializes by requesting the current analysis state from the background worker via chrome.runtime.sendMessage. The received state is used to populate all three display panels: the summary panel, the Chart.js visualization panel, and the comment browser panel.

Chart.js is initialized with a doughnut chart configuration mapping each of the seven signal categories to a distinct color. The chart dataset is updated reactively when new analysis results arrive. The comment browser panel renders an HTML list of comment entries, each annotated with a category badge and a confidence percentage derived from the model's output probability for the predicted class. Category filter buttons above the list toggle a CSS filter class that shows or hides comment entries matching the selected category.

8.4 Flask Backend Implementation

8.4.1 Application Structure

The Flask application is structured as a single Python module (app.py) with supporting utility functions for preprocessing and model loading. The application initializes the RoBERTa tokenizer and model at startup using the Hugging Face AutoTokenizer and AutoModelForSequenceClassification classes, loading the fine-tuned model checkpoint from the local model cache or from the Hugging Face Hub if not already cached. The model is moved to the available device (CPU in the standard deployment configuration) and set to evaluation mode.

8.4.2 Inference Endpoint

The /classify endpoint handler (classify_comments function) performs the following operations for each incoming request: it validates the request JSON structure, extracts the comments array, applies the preprocessing pipeline to each comment, tokenizes the preprocessed comments in mini-batches using the RoBERTa tokenizer's batch_encode_plus method with padding and truncation enabled, passes each mini-batch through the model within a torch.no_grad() context, applies softmax to the logit outputs to obtain class probability distributions, and constructs the response JSON with predicted labels and probability dictionaries.

8.4.3 CORS and Error Handling

Flask-CORS is applied to the entire application using the CORS(app) decorator, allowing cross-origin requests from any origin (required for extension background worker access from the chrome-extension:// scheme). Custom error handlers are registered for HTTP 400 (malformed request), HTTP 500 (internal server error), and connection timeout scenarios. All error responses follow a consistent JSON format with an error code and a human-readable message.

8.5 Website Frontend Implementation

The website frontend at https://signal-nine-phi.vercel.app/ is implemented as a modern web application maintained in the repository at https://github.com/achhajer/mental-health-analyzer. The website provides a clean, accessible interface with a prominent text input area accepting free-form text for analysis, a submit button that triggers the inference request, and a results panel that displays the detected signal categories and their associated confidence scores upon completion.

The website connects to the RoBERTa-based inference backend using the Hugging Face Inference API or a compatible locally hosted endpoint. The frontend handles loading states with appropriate visual feedback, displays error messages for network failures or backend unavailability, and presents results in a structured, user-friendly format that reflects the project's responsible AI framing: outputs are labeled as signal indicators rather than diagnoses.

Deployment is handled through Vercel's GitHub integration. The Vercel project is linked to the website repository, and each merge to the main branch triggers an automatic build and deployment. The deployed site is served over HTTPS with global CDN distribution, ensuring low latency for users worldwide.

8.6 Version 2 Implementation

8.6.1 Dataset Redesign and LLM-Assisted Annotation

The Version 2 dataset is constructed using a two-stage process. In the first stage, text samples are collected from multiple Reddit communities using an expanded query strategy that goes beyond simple subreddit membership. In the second stage, a locally deployed large language model (via Ollama or an equivalent local inference framework) is used to review each collected sample and assign structured multi-label annotations based on explicit signal criteria defined by the project team.

The annotation prompt instructs the LLM to evaluate each text sample along each of the V2 signal dimensions and to assign binary presence/absence labels as well as a three-point severity estimate (low, moderate, high) for each present signal. This annotation methodology produces labels that are significantly more semantically grounded than subreddit-based weak supervision, reducing the label noise that was a recognized limitation of V1 [11, 12].

8.6.2 Multi-Task RoBERTa Model

The V2 model architecture extends RoBERTa-base with a multi-task classification head. For each signal dimension, a separate binary classification head (consisting of a linear layer with sigmoid activation) is attached to the pooled [CLS] representation. An additional shared severity prediction head outputs a three-class ordinal estimate for the aggregate distress signal of the input. The entire multi-task model is trained jointly using a weighted sum of binary cross-entropy losses for the signal heads and a cross-entropy loss for the severity head.

8.6.3 Knowledge Distillation Pipeline

Following teacher model training, the knowledge distillation pipeline trains a more compact student model by minimizing a combined loss function that includes both the task-specific ground-truth losses and a distillation loss measuring the Kullback-Leibler divergence between the teacher's soft output distributions and the student's outputs [5]. The temperature parameter controlling the softness of the teacher's distributions is set to 4.0, which has been empirically shown to produce effective distillation in sequence classification tasks.

8.6.4 ONNX Conversion and INT8 Quantization

After distillation training, the student model is exported to the ONNX format using PyTorch's torch.onnx.export function with dynamic axes configured for variable sequence length and batch size. The exported ONNX model is then subjected to post-training static INT8 quantization using the onnxruntime.quantization module. A calibration dataset consisting of a representative sample of the training corpus is used to determine the optimal quantization parameters for each operator in the model graph.

The quantized model is validated against the original student model's predictions on the validation set to confirm that quantization-induced accuracy degradation is within acceptable bounds (empirically observed at less than 1.5 percentage points on the macro F1-score). The final quantized ONNX model achieves a storage size of approximately 125 MB, compared to approximately 499 MB for the full V1 RoBERTa model, representing a reduction of approximately 75%.

8.6.5 SQLite Wellbeing Store Implementation

The optional SQLite wellbeing store is implemented as a simple database with a single session_records table containing columns for the session timestamp, the page URL analyzed, the detected signal distribution as a JSON blob, the aggregate severity category, and the session duration in seconds. The extension writes to this database via a small Python service exposed on localhost, ensuring that all storage operations remain local. The wellbeing dashboard reads from this store to produce trend visualizations covering the user's analysis history.

Chapter 9: Results and Discussion

This chapter presents the quantitative and qualitative results of the Mental Health Signal Analysis System evaluation. The primary evaluation is conducted on the Version 1 fine-tuned RoBERTa classifier. Results are compared against three baseline systems and analyzed both at the aggregate level and at the per-class level. The chapter also discusses the V2 deployment improvements and interprets the results in the context of the project's goals.

9.1 Version 1 Evaluation: Aggregate Performance

The fine-tuned RoBERTa classifier for seven-class mental health signal analysis was evaluated on a held-out test set comprising 20% of the total dataset, stratified by class to maintain category proportions. The primary evaluation metrics are accuracy (proportion of correctly classified samples), macro-averaged F1-score (unweighted mean of class-wise F1-scores), and weighted F1-score (F1-score weighted by class support).

The fine-tuned RoBERTa model achieved an accuracy of 0.8569 (85.69%) and a macro F1-score of 0.8601 on the test set. These results represent a substantial improvement over all three comparison baselines. The weighted F1-score of 0.8569 confirms that the model's strong macro performance is not an artifact of class imbalance.

Table 5.1: Comparative Performance of Classification Models on Seven-Class Mental Health Dataset

Model	Accuracy	Macro F1
Fine-Tuned RoBERTa (Proposed)	0.8569	0.8601
BERT-base (Fine-Tuned)	0.7950	—
SVM + TF-IDF (Logistic Regression)	0.6815	—
Lexicon / Dictionary-Based	0.3800	—

[Figure 9.1: Comparative Bar Chart — Model Accuracy vs Classification Approach — insert chart here]

9.2 Version 1 Evaluation: Per-Class Performance

The class-wise performance of the fine-tuned RoBERTa model is presented in the table below. The results reveal a consistent pattern: categories that are linguistically more distinct and semantically more coherent (Autism, ADHD, and Casual Conversation) achieve higher precision and recall, while categories with more overlapping linguistic profiles (Depression and BPD) show somewhat lower scores. This is a predictable and well-documented characteristic of mental health NLP datasets, where disorders that share clinical features (such as depression and BPD, which both frequently involve expressions of hopelessness and emotional dysregulation) also share surface-level linguistic patterns.

Table 5.2: Class-wise Precision, Recall, and F1-Score for Fine-Tuned RoBERTa (Version 1)

Category	Precision	Recall	F1-Score
ADHD	0.8989	0.8959	0.8974
Anxiety	0.7885	0.8301	0.8087
Autism	0.9082	0.8813	0.8946
BPD	0.8278	0.7867	0.8067
Depression	0.7729	0.7811	0.7769
PTSD	0.8646	0.8606	0.8626
Casual Conversations	0.9638	0.9842	0.9739
Macro Average	—	—	**0.8601**

[Figure 9.2: Per-Class F1-Score Distribution for Fine-Tuned RoBERTa — insert bar chart here]

9.3 Discussion of Results

9.3.1 Transformer Superiority over Classical Methods

The results in Table 5.1 confirm the now well-established finding that contextual transformer representations substantially outperform classical feature-engineering approaches for mental health text classification. The fine-tuned RoBERTa model outperforms the lexicon-based baseline by 47.69 percentage points in accuracy, and outperforms the SVM/TF-IDF approach by 17.54 percentage points. This gap reflects the fundamental advantage of transformer-based contextual embeddings: rather than treating each word as an independent feature, RoBERTa represents each token in the context of its surrounding tokens, capturing the nuanced, context-dependent meaning of language that is essential for accurate mental health signal detection.

9.3.2 RoBERTa vs BERT-base

The fine-tuned RoBERTa model outperforms the fine-tuned BERT-base model by 6.19 percentage points in accuracy. This improvement is consistent with the theoretical advantages of RoBERTa's training methodology documented in [3]: larger pre-training batches, dynamic masking, and the removal of the next-sentence prediction objective collectively produce richer contextual representations that transfer more effectively to fine-tuning tasks. The magnitude of the improvement on this specific domain confirms that RoBERTa's advantages generalize to social media mental health text.

9.3.3 Per-Class Analysis

The Casual Conversations class achieves the highest F1-score (0.9739), which is expected given that this class represents general conversational text that is linguistically distinct from the disorder-associated categories. The ADHD class (F1: 0.8974) and Autism class (F1: 0.8946) also achieve strong scores, likely because these communities on Reddit discuss very specific and distinctive aspects of their experiences (executive function challenges, sensory processing, social interaction patterns) that provide clear linguistic markers.

Depression (F1: 0.7769) and BPD (F1: 0.8067) achieve the lowest F1-scores among the disorder categories. This is consistent with the documented clinical and linguistic overlap between these conditions: both frequently involve expressions of emotional pain, relationship difficulties, and hopelessness, making them harder to distinguish from each other and from general expressions of distress. This observation directly motivates the V2 shift to multi-label analysis, where a comment expressing both depressive and BPD-associated signals can legitimately receive both labels rather than being forced into one category.

9.4 Version 2 Deployment Improvements

The Version 2 deployment pipeline introduces measurable improvements in deployment efficiency while maintaining model quality. The table below summarizes the key deployment characteristics of V1 versus V2.

Table: Version 1 vs Version 2 Deployment Characteristics

Property	Version 1	Version 2
Model Type	Single-label RoBERTa	Multi-task RoBERTa (teacher–student)
Deployment Format	PyTorch (.bin)	ONNX + INT8 Quantized
Deployment Size	~499 MB	~125 MB
Output Type	One dominant class	Multi-label signals + severity
Local Persistence	None	SQLite Wellbeing Store
User Surface	Extension only	Extension + Website

Annotation Method	Subreddit-based (weak)	LLM-assisted structured annotation

Chapter 10: Ethical Considerations and Privacy

The application of machine learning to mental health-related language carries inherent ethical risks that require careful consideration and deliberate design choices. This chapter addresses the ethical dimensions of the Mental Health Signal Analysis System, covering privacy protection, the risk of diagnostic overclaiming, data ethics, system transparency, and the responsible design of user-facing features.

10.1 The Privacy-First Architecture

The most fundamental ethical design decision in this project is the commitment to local-first inference. All comment processing, tokenization, model inference, and result storage occurs exclusively on the user's local machine. No comment text, user identifiers, or analysis results are transmitted to any external server operated by the project team or by any third party.

This design is grounded in the privacy-preserving NLP literature [15], which identifies local inference as the strongest available privacy protection for sensitive language data. Mental health-related text is among the most sensitive categories of personal data: it can reveal psychological vulnerabilities, stigmatized conditions, and deeply personal experiences. The consequences of such data being exposed, intercepted, or misused could be severe for the individuals involved. By eliminating network transmission of this data entirely, the project provides a guarantee of privacy that no server-side system can match.

The Hugging Face model hosting is used only for model checkpoint distribution: the model weights are downloaded once to the user's local machine and thereafter accessed entirely locally. The Vercel-hosted website does connect to inference endpoints for its text analysis feature, but this connection is made explicit to users and the submitted text is processed transiently without logging or storage.

10.2 Responsible Framing: Signal Analysis vs Diagnosis

One of the most consequential ethical choices in this project is the deliberate avoidance of diagnostic language. Phrases such as "this user has depression" or "this comment indicates a diagnosis of PTSD" are never used by the system. Instead, all outputs are framed as signal indicators, pattern observations, or probabilistic estimates.

This distinction is not merely semantic. The ethical literature [16] identifies several specific harms that can arise from framing NLP outputs as diagnoses: stigmatization of individuals based on automated classification of their public statements; potential legal liability if diagnostic claims prove incorrect; inappropriate self-diagnosis or anxiety in users who read their browsing environment results as a medical assessment; and the creation of a false authority relationship between the system and the user.

The project addresses these risks by consistently using the terminology of signal analysis throughout the interface, documentation, and report. The transition from Version 1's language of "disorder detection" to Version 2's language of "signal analysis" is itself an ethical improvement, reflecting the project team's critical engagement with the responsible AI literature during the evolution of the system.

10.3 Data Ethics and Informed Consent

The training data for the V1 model was collected from public Reddit posts. While these posts are publicly accessible, the authors of those posts did not explicitly consent to their use as training data for a mental health classifier. This is a recognized ethical tension in social media NLP research [16] that the project acknowledges openly.

Standard practices for mitigating this concern in the research community include the use of aggregate rather than individual-level analysis, the avoidance of re-identifying individuals from model outputs, and the framing of collected data as reflective of population-level language patterns rather than as records of specific individuals. The project adheres to all of these practices. No individual's Reddit identity is stored or processed beyond the training phase, and model outputs are presented as community-level signal patterns rather than individual profiles.

10.4 Fairness and Bias Considerations

Mental health NLP systems trained on social media data are susceptible to biases arising from the demographic characteristics of social media users, the cultural specificity of mental health expression, and the systematic differences in how people from different backgrounds discuss mental health online. A classifier trained predominantly on English-language, Western social media text may underperform or produce biased outputs for users from other linguistic or cultural contexts.

The project acknowledges this limitation explicitly and identifies multilingual support and cross-cultural validation as important directions for future development. The current system is designed for English-language text and does not make claims about its performance or appropriateness for other language communities. This transparency is itself an ethical practice: users are made aware of the system's scope and limitations rather than having those limitations obscured.

10.5 Responsible Design of User-Facing Features

The Version 2 wellbeing features — Smart Shield, Rabbit-Hole Nudges, Emotional Diet, and Draft-Time Support Prompts — are designed in accordance with the responsible interface design principles discussed in [17]. Each feature is explicitly opt-in: the system does not activate any intervention without the user's explicit configuration. The language used in prompts and notifications is calm, non-alarmist, and non-judgmental. No feature claims to provide mental health advice or to replace professional support.

The Rabbit-Hole Nudge, for example, does not tell a user that they are in distress or that they should stop using a platform. It simply surfaces a brief awareness prompt noting that the user has been browsing in a high-distress signal environment and offering them a moment of reflection. This framing respects the user's autonomy while providing potentially meaningful support, consistent with the ethical design literature [17].

10.6 Transparency and Explainability

All outputs of the system are accompanied by explicit explanations of what they represent: probability scores from a text classification model trained on social media data. The extension popup includes a brief explanatory text reminding users that results are signal estimates and not clinical assessments. The website includes a similar disclaimer prominently on the results page.

The project repositories are publicly accessible on GitHub, allowing any interested user or researcher to inspect the system's source code, model training scripts, and data preprocessing pipeline. This openness supports reproducibility and independent verification of the system's claimed properties, both of which are important components of responsible AI practice.

Chapter 11: Future Scope

While the Mental Health Signal Analysis System represents a significant achievement as a complete, deployed, and evaluated NLP platform, a number of important extensions and improvements are identified as future scope. These directions are grounded in the limitations of the current system and in the broader research literature surveyed in Chapter 4.

11.1 Multilingual Support

The current system is designed exclusively for English-language text. Mental health discussions are conducted in every language on social media platforms, and a system limited to English excludes the majority of the world's social media users. Future work should explore multilingual pre-trained models such as XLM-RoBERTa as backbone architectures, enabling the system to analyze comments in multiple languages within a single unified model. Cross-cultural validation of the signal categories would also be required, as mental health expression varies substantially across cultural contexts.

11.2 Broader Platform Support

The Version 1 extension currently supports YouTube, Reddit, and X/Twitter. Expanding support to additional platforms such as Facebook, Instagram, LinkedIn, TikTok, and Discord would significantly increase the system's utility. Each new platform requires the development of platform-specific content extraction logic, which may need to be updated periodically as platform DOM structures evolve. A plugin architecture for the content script layer would facilitate more maintainable addition of new platform support.

11.3 Multimodal Analysis

The current system analyzes text exclusively. Social media content increasingly combines text with images, audio, and video. Future extensions of the system could incorporate multimodal analysis, using vision-language models to analyze image content or audio transcription to extend signal detection to spoken content in short-form videos. This would significantly expand the breadth of signal that the system can detect but would also introduce additional ethical complexity around image and audio privacy.

11.4 Longitudinal Analytics and Personalized Insights

The V2 SQLite wellbeing store provides the foundation for longitudinal trend analysis, but the current implementation captures only session-level aggregates. Future development could extend this to richer longitudinal analytics, including tracking how the signal character of a user's browsing environment changes over days, weeks, and months. Personalized insights derived from this longitudinal data — presented with appropriate caveats and in consultation with mental health guidelines — could provide meaningful awareness support for users who choose to engage with this feature.

11.5 Clinical Collaboration and Expert-in-the-Loop Review

The current system operates entirely without clinical oversight. A meaningful future direction would be to engage mental health professionals in the validation of the system's signal taxonomy, the calibration of its severity estimates, and the design of its user-facing intervention features. Expert-in-the-loop review of the model's outputs on difficult or ambiguous cases would improve the system's reliability and would provide an important safeguard against systematic errors in the automated classification pipeline.

11.6 Federated Learning and Differential Privacy

The current local-inference architecture provides strong privacy protection but does not allow the system to improve its models from aggregated user experience without centralizing user data. Federated learning techniques, in which model updates are computed locally and only gradient information (not raw data) is shared with a central aggregation server, could enable continuous model improvement while preserving user privacy. Differential privacy mechanisms could further strengthen the privacy guarantees of such a federated training pipeline. These techniques are discussed in the privacy-preserving NLP literature [15] as important directions for future systems in sensitive domains.

11.7 Mobile Application

The current system is limited to the desktop Chrome browser environment. A mobile application for iOS and Android platforms would significantly expand the system's reach, as social media consumption increasingly occurs on mobile devices. Developing a mobile version would require adapting the comment extraction logic to mobile browser environments or native app integrations, and would need to address the computational constraints of mobile hardware for local model inference.

11.8 Stronger Evaluation with Clinical Ground Truth

The V1 model is evaluated on a held-out portion of the same weakly supervised dataset used for training. While this evaluation is informative, it does not assess the model's performance against clinical ground truth labels. A stronger future evaluation would involve collecting a test set annotated by qualified mental health clinicians, providing a more rigorous assessment of the system's real-world signal detection accuracy. Collaboration with research institutions conducting mental health studies would facilitate such an evaluation.

Chapter 12: Conclusion

This report has presented the Mental Health Signal Analysis System, a privacy-preserving natural language processing platform developed as a final year project at Heritage Institute of Technology, Kolkata. The system was designed, implemented, and evaluated as a coherent two-version evolution: Version 1 as a fully deployed, locally executing Chrome browser extension with a complementary website interface, and Version 2 as a methodologically improved redesign introducing multi-label signal analysis, severity estimation, knowledge distillation, and ONNX quantization.

Version 1 demonstrated that fine-tuned transformer models are substantially superior to classical NLP approaches for mental health signal classification on social media text. The fine-tuned RoBERTa model achieved 85.69% accuracy and a macro F1-score of 0.8601, outperforming BERT-base by 6.19 percentage points, SVM/TF-IDF by 17.54 percentage points, and lexicon-based methods by 47.69 percentage points. These results provide a strong empirical foundation for the project's technology choices and confirm the effectiveness of contextual transformer representations for this domain.

The critical analysis underlying Version 2 demonstrated the project team's capacity for genuine research maturity: rather than treating Version 1's strong results as a final answer, the team identified the conceptual limitations of single-label disorder classification, the ethical inadequacy of subreddit-based weak supervision, and the deployment inefficiency of full-scale model weights. Version 2's responses to each of these limitations — multi-label signal analysis, LLM-assisted annotation, and ONNX quantization — are each grounded in established research literature and represent principled methodological improvements rather than arbitrary feature additions.

The project's commitment to privacy-first design is reflected in every architectural decision: all inference occurs locally, no user data is transmitted to external servers, and the optional V2 wellbeing store operates entirely on the user's device. The responsible framing of all outputs as signal indicators rather than diagnoses reflects the team's engagement with the ethics literature and demonstrates an awareness of the real-world consequences of overclaiming in sensitive AI applications.

The system is accessible through two complementary user surfaces: the Chrome browser extension at https://github.com/Ekam-Bitt/Detection-of-Mental-Disorders-Extension, which provides live in-context comment analysis during active browsing, and the website frontend at https://signal-nine-phi.vercel.app/, which provides direct text analysis in a clean, standalone interface. Together, these surfaces bring the capabilities of advanced mental health NLP to ordinary users in a practical, privacy-respecting form that represents a genuine contribution beyond the standard academic prototype.

In summary, this project contributes a fully implemented, locally deployed, ethically designed mental health signal analysis platform that demonstrates both the practical engineering competence and the research maturity of the project team. It begins with a working product, evaluates it rigorously, identifies its limitations honestly, and proposes and partially implements a principled redesign. This is the story of a project that not only built something that works, but asked why it works, where it falls short, and how it should be improved.

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